



AMRITA VISHWA VIDYAPEETHAM

**AMRITA SCHOOL OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING**

Amritanagar, Coimbatore - 641112

ROBOTICS & CONTROL LAB MANUAL

DEPARTMENT OF MECHANICAL ENGINEERING

LIST OF EXPERIMENTS

Expt. No	Experiments
	PART – A
1	Introduction to Mitsubishi Industrial Robot
2	Palletizing operation.
3	Stacking operation
4	Determining Tool Center Point through software
5	Draw figure 8 using offset
6	Define a virtual fence for the robot
7	Identify singular configurations of the robot
8	Vision-Based Shape Detection and Sorting Robot
9	Vision-Based Colour Sorting
	PART – B
10	Work Envelope and Configuration of a 2R Robotic Arm
11	Inverse kinematics of a 2R Robotic Arm
12	Forward and inverse kinematics of 3R Robotic Arm
13	Introduction to the Robotics Systems Toolbox (RST) in MATLAB
14	Forward and Inverse Kinematics of robotic arm using RST
15	Forward and Inverse Dynamics of robotic arm using RST
16	Trajectory generation for the robotic arm using RST
17	PID based Trajectory Tracking Control of a 2R Robotic Arm